Statistics and Trends

Student Mathematics Performance Analysis

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Github Link : Github Link

1. Introduction: Academic performance is important for understanding the progress of students. The grades in mathematics provide relevant information regarding students' academic performance, specifically the correlation between first-period and final grades and the effect of the previous class's failures. These results help to identify differences in students' performance in math.

2. Data Description:

Table 1 : Dataset Overview and Selected Variables

Description	Details
Dataset name	student-mat.csv
G1	First-period grade
G2	Second-period grade
G3	Final mathematics grade
studytime	Weekly study-time category
failures	Number of previous class failures
Data types	Numerical and categorical

3. Statistical Analysis and Visualisation:

3.1 Histogram – Distribution of Final Mathematics Grades (G3)

Figure 1 shows the histogram of the final mathematics grade for 395 students. Most students scored between 8 and 16, with the highest frequency between 10 and 12. The mean grade is 10.42, the median is 11 and the standard deviation is 4.58. So fewer students got the higher grades, and some got zero. So there is variation in the maths.

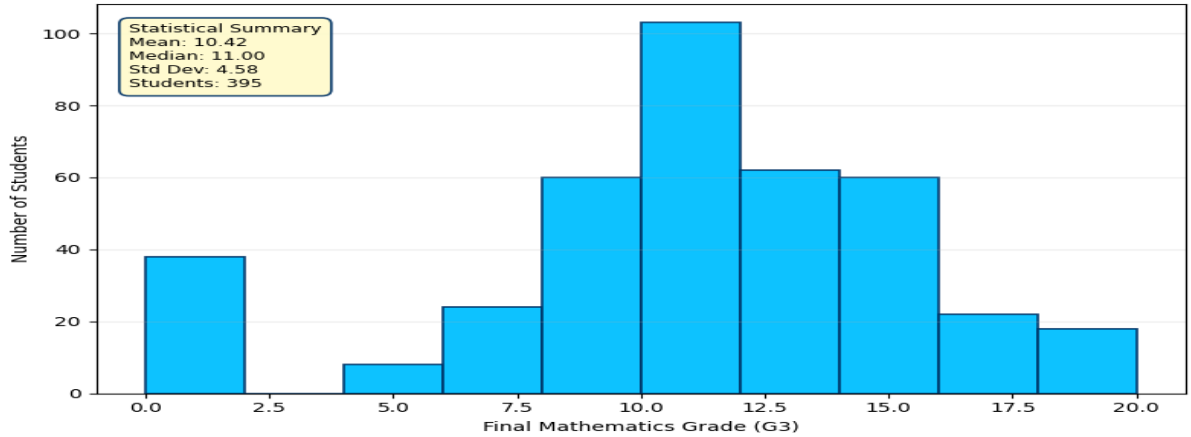


Figure 1: Distribution of Final Mathematics Grades (G3).

3.2 Scatter Plot – Relationship Between First-Period and Final Mathematics Grades

Figure 2 shows the relationship between first period and final grades in mathematics in the scatter plot. The points seem to trend upwards from left to right. The two variables are strongly positively related with a correlation coefficient of 0.8015. Students who score higher in G1 are likely to score higher in G3.

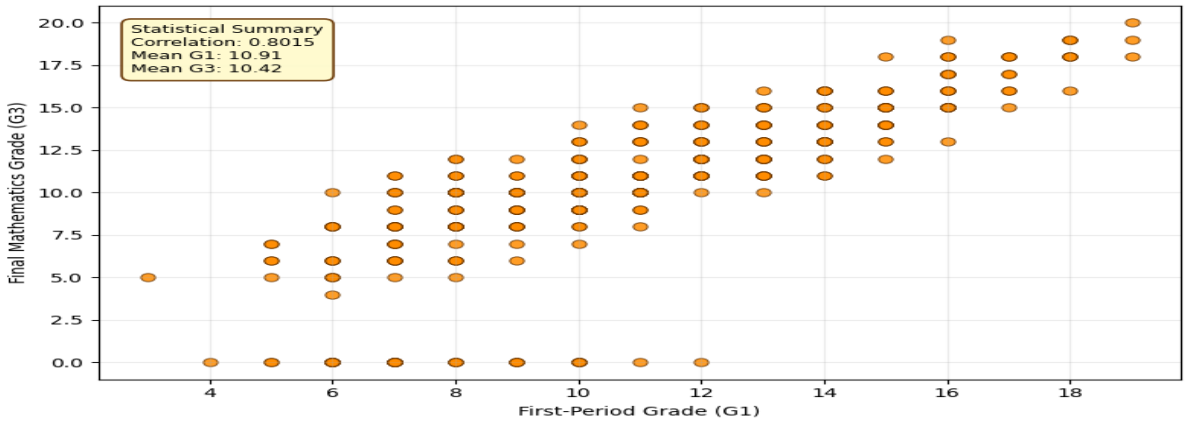


Figure 2: Relationship Between First-Period and Final Mathematics Grades (G1 vs G3)

3.3 Box Plot – Final Mathematics Grades by Previous Class Failures

As shown in Figure 3, box plot compares students' final grades according to previous class failures. Students who have not previously failed tend to have a higher median grade. The median grade tends to decline with an increase in the number of previous failures. Students with more previous failures generally show lower final grades.

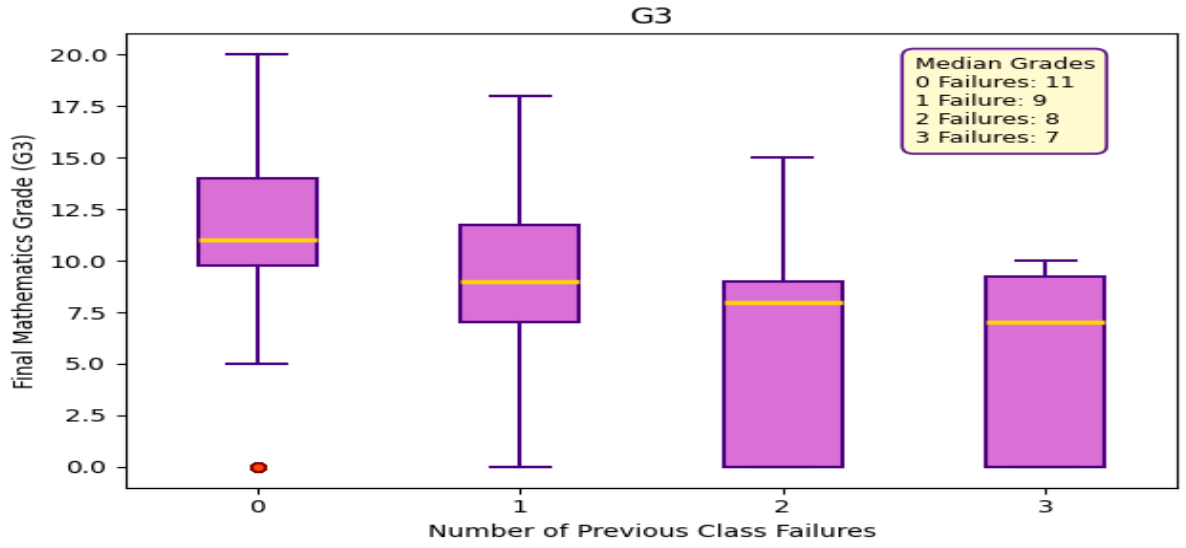


Figure 3: Final Mathematics Grades by Previous Class Failures (G3 vs failures)

4.Conclusion : The three plots provide useful insights into students' mathematics performance. They show the distribution of grades, the relation between G1 and G3 and the effect of previous failures in the class.